

The Corrosive Nature of Diluted Bitumen and Crude Oil

Literature Review

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Executive Summary

Crude oil, in particular heavy and extra-heavy crude, is both corrosive and toxic due to its contaminants. This is a fact unquestioned by academia, government and industry. The corrosiveness of crude oil and diluted bitumen creates hazards for pipelining, storage tanks and shipping. The degree of corrosion and its risk management depend on the levels of contaminants found in crude oil or diluted bitumen. The following aspects of crude oil and diluted bitumen suggest that they both pose considerable corrosion risk:

1. It is accepted in both science and industry that crude oil is corrosive: ASTM's standard G205-10 was written solely as a guide for determining the level of crude oil corrosiveness. The National Research Council's website on pipeline integrity states that internal corrosion caused the majority (about 50%) of pipeline failures, including those carrying crude oil.
2. Diluted bitumen is at the contaminant-heavy end of the crude oil scale and has the potential to create serious corrosion problems for pipelines, storage tanks and cargo holds, and to cause toxic effects in the environment.
3. Current standards and codes do not distinguish between diluted bitumen and conventional crude oil.
4. The level of corrosiveness is primarily dependent upon the contaminant concentration in petroleum products. Bitumen and diluted bitumen (dilbit) have higher levels of sulphur, heavy metals, TANs and asphaltenes than conventional crude oil.
5. Sulphur is corrosive to carbon steel which is the material used for pipelines, storage tanks and cargo holds. Higher concentrations of sulphur pose greater threats to metal integrity. Corrosion in pipelines, storage tanks and cargo holds is known to occur from sulphur-reducing bacteria, H_2S , or atomic sulphur.
6. Literature cites average sulphur concentrations of 3.7% in diluted bitumen (bitumen concentrations are between 4% and 7%). The heaviest sour crude oils contain less than 3%.
7. Stress corrosion cracking (SCC) can develop once sulphur has created a corrosive environment within a pipeline, storage tank or cargo hold, thereby increasing the risk of failure.
8. Heavy metals such as nickel and vanadium are contaminants in crude oils and are more highly concentrated in diluted bitumen than in conventional crude oil. They are known to be highly toxic in the environment.
9. Asphaltenes are more highly concentrated in diluted bitumen than conventional crudes and are known to cause sedimentation and sludge by precipitating out. The development of particulates depends on the characteristics as of the particular diluent being mixed with the bitumen. The diluent (typically condensate) used to thin the bitumen can vary on a daily basis and therefore the amount of particulate can also vary on a daily basis. Diluent composition depends on the source and process from which it originates and therefore its corrosiveness and toxicity vary as well.

10. The 2006 crude oil pipeline rupture in Alaska was due to internal corrosion with evidence of sulphur-reducing bacteria. Areas of the pipeline shown to be free of corrosion one year before the spill, showed high rates of corrosion at the time of the spill.
11. The PHMSA Significant Incidents Listing website lists six Enbridge pipeline spills since 2004 due specifically to internal corrosion; volume from these spills totalled 111,610 L.
12. Enbridge's 2010 diluted bitumen rupture in Michigan was from a pipeline along which evidence of corrosion had been noted prior to the spill. A hydrogen sulphide warning was issued to the public after the spill.

Without a more thorough understanding of the corrosive effects of diluted bitumen during pipelining and storage, it is not possible to assess the risks posed by transporting and storing diluted bitumen. As shown by the evidence after the Prudhoe Bay rupture, rapid-onset corrosion can occur and proceed at a high rate. More studies which provide evidence of successful design are required before the risk might be considered low enough to build new pipelines.

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1. Introduction

Chemical and physical characterisations of diluted bitumen and condensate are essential for determining and understanding the implications on the pipelining and storage of these petrochemicals. This report serves as a literature review on the classification of bitumen and on the corrosive nature of all crudes. The corrosiveness presents hazards to pipelining, storage tanks and shipping; should corrosion result in a rupture, the toxicity of crudes, in particular of diluted bitumen, will result in severe environmental damage.

2. Similarities and Differences between Conventional Crude and Bitumen

2.1 Diluted Bitumen and Heavy Crude Oil

Crude oil, as defined by the International Energy Association (IEA), “exists in the liquid phase under normal surface temperature and pressure and usually flows to the surface under the pressure of the reservoir.”

Bitumen, also known as heavy oil or oil sands, is formed by the microbial degradation of conventional crude oils. This process significantly alters the chemical composition as well as the fluid properties of the oil. It is one of the most complex molecules found in nature and has hydrocarbon chains in excess of two thousand molecules (Gov’t of Alberta, 2009). The majority of bitumen is C_{30+} material (Brierley *et al.*, 2006).

Petroleum products such as bitumen and crudes are typically characterised by their American Petroleum Institute (API) gravity, which is a measure of how heavy or light a liquid is when compared to water. If oil’s API gravity is greater than 10° , then it will float on water; if it is less than 10° , it is heavier than water and will sink. The API of bitumen is between 8° and 14° (Gov’t of Alberta, 2009). Heavy crude oil, as defined by IEA, has an API less than 20° . In Boduszynski *et al.* (1998), the ranking of “heavy” or “light” in crudes is also based on density and it lists heavy crudes as having an API range of 10° to 20° . In this ranking, bitumen is considered either extra-heavy crude or simply “bitumen”. Due to the lower API and higher viscosity of bitumen, it must be diluted for transport.

The IEA distinguishes between bitumen and conventional crude oil for reasons other than their API; it categorizes the former under the heading “Unconventional Oils” since any oil obtained by unconventional production techniques is labelled thus (IEA website). The Canadian Association of Petroleum Producers (CAPP), in its *Crude Oil: Forecast, Markets & Pipelines* (June, 2011), also distinguishes between conventional and oil sands crudes.

“The CAPP forecast categorizes the various crude oil types that comprise western Canadian crude oil supply into four major categories: Conventional Light, Conventional Heavy, Upgraded Light and Oil Sands Heavy. Oil Sands Heavy includes upgraded heavy sour crude oil, bitumen diluted with upgraded light crude oil (also known as “SynBit”) and bitumen diluted with condensate (also known as “DilBit”). An example of such a DilBit would be Cold Lake crude, which has a density of about 930 kg/m^3 (21° API^*) and a sulphur content of 3.6 per cent.”

CAPP, 2011

*dilbit has a higher API than bitumen because it has already been diluted and is therefore thinner

Industry, too, distinguishes bitumen from conventional crude. As stated by Bruce (2005) in Jacobs Canada Inc.'s presentation at a Canadian Heavy Oil Association Technical Luncheon, "The bitumen portion is "nasty"". He mentioned the challenging chemistry as defined by bitumen containing significant aromatic and asphaltenes compounds, high sulphur, nitrogen and metals, and stated that bitumen "also contains highly corrosive organic acids". The industry journal *Oil Sands Review* has a glossary on its website (www.oilsandsreview.com) which defines bitumen as "(a) naturally occurring, viscous mixture of hydrocarbons that contains high levels of sulphur and nitrogen compounds."

However, because of the relative novelty of pipelining diluted bitumen (as compared to conventional crude oil), the US Pipeline and Hazardous Materials Safety Administration (PHMSA) does not distinguish between conventional crude and diluted bitumen when setting standards. Correspondence with the Canadian Standards Association regarding CSA standard Z662-07* reveals that CSA also does not yet distinguish between the two commodities. An email response dated January 26, 2011 states:

"... The term "bitumen" is not used anywhere in Z662. This Standard covers the design, construction, operation, and maintenance of oil and gas industry pipeline systems that convey (a) liquid hydrocarbons, including crude oil, multiphase fluids, condensate, liquid petroleum products,...".

Canadian Standards Association, 2011

*the email was sent prior to the release of CSA Z662-11. Further correspondence on August 2, 2011 confirmed that the updated version still does not refer to "bitumen" or "diluted bitumen".

2.2 Diluents and Diluted Bitumen

Bitumen on its own is too thick to flow, so it must be diluted before pipelining is viable. Diluent is defined by the CAPP as "hydrocarbon blended with heavy crudes and or bitumens to enable the meeting of pipeline viscosity and density targets" (CAPP, 2008). There are various diluents available for bitumen, the most common being synthetic oil and condensate. In the oil sands industry, the terms dilbit and synbit are used to describe the blends created by using condensate and synthetic oil as diluents. Because natural gas condensate is in short supply in regions where bitumen is found, producers have presented synthetic oil as an alternative.

Dilbit is blended bitumen, using condensate as its diluent; it is used when the required diluent density is less than 800 kg/m^3 and typically blends at a 25/75 ratio (CAPP, 2008). Synbit is blended bitumen which uses synthetic crude oil as its diluent and is used when the diluent density should be greater than 800 kg/m^3 ; it typically blends at a 50/50 ratio (CAPP, 2008). It is because synthetic crudes have a much lower API gravity than condensate (meaning they are more viscous), that more diluent is required to thin out the bitumen.

The choice of diluent is a question of availability and cost. Condensate is less expensive yet in short supply. Synthetic crudes contain no residue while dilbits have both high residual content and high sulphur content (Brierley *et al.*, 2006). It is generally accepted in industry that since condensate supplies are limited, synthetic crude will be used more and more in the future (Crandall, 2002).

A joint 2010 presentation by the Enbridge CRW Committee (CRW is the industry moniker for condensate blend) and CAPP stated that there are two principal selection components when condensate is to be used as a diluent: performance characteristics as determined by API gravity/viscosity and risks posed by the condensate's characteristics.

A condensate's risks are due both to its physical and chemical properties. Sediment is found in condensate and thus sedimentation may occur. This can lead to the deposition of solids in low flow zones of a pipeline and in the bottom of storage tanks, or erosion of interior pipeline metal through scraping action. Both oil and condensate extracted from a reservoir contain water and sediment, most of which is extracted. If the residual content of these unwanted impurities is too high, it can lead to sediment deposits. The chemical properties of condensate which may pose risks include (but are not limited to) aromatics, olefins, organic chlorides and sulphur compounds.

2.2 Sulphur Content

Crudes are also categorized in terms of their sulphur content, with sweet crudes having less than 0.5% sulphur content and sour crudes having greater than 0.5% (IEA definition). Certain components of the sulphur within bitumen and sour crudes are corrosive and therefore weaken the integrity of pipelines and storage tanks (see Section 3).

Vathyam and Lee (2009) list the concentration of sulphur in Alberta bitumen as "high", varying from 4.0% to 6.7%, with an average of 5.2% (recall that to be termed "sour", the crude must only contain 0.5% sulphur or greater). Crandall's presentation to the International Energy Agency conference in 2002 confirms this, saying that the sulphur content of bitumen is high. Environment Canada's Oil Properties Database lists Athabasca Bitumen as containing a sulphur range of 4.41% to 5.44%. The Government of Alberta's Oil Sands Discovery Centre fact sheet on bitumen says that bitumen, on average, contains 4.8% sulphur (Gov't of Alberta, 2009). Enbridge's own website lists the content in the mid-3% range.

Conventional crudes have lower sulphur content, even those classified as sour. Environment Canada's Oil Properties database lists Alaska North Slope crude as ranging from 1.14% to 1.39% sulphur and Arabian Heavy as having 2.8% sulphur. The 2010 Baker Hughes white paper *Planning Ahead for Effective Canadian Crude Processing* compares Athabasca Bitumen to Arab Heavy, listing sulphur as 4.9% for bitumen and 2.8% for Arab Heavy. In another industry contribution, Purvin & Gertz's 2002 presentation to the International Energy Agency Conference on Non-Conventional Oil, states that bitumen and extra-heavy crude oil have high sulphur content and high residual content; many also have high TAN content (see Section 2.4). The presentation lists a sulphur content of 4.5% for Athabasca bitumen and 3.7% for Athabasca dilbit (Crandall, 2002).

While the sulphur content in bitumen is high, it may be assumed that upon blending with diluent (which has low sulphur content), the concentration would decrease substantially but this does not prove to be the case. Crandall (2002) states that Athabasca Dilbit and SynBit have 3.7% sulphur and 2.8% sulphur, respectively. In another industry presentation, Jacobs Canada shows a sulphur concentration of 3.79% in dilbit as compared to West Texas Intermediate with 0.44% sulphur (Bruce, 2005).

The primary concern with sulphur corrosiveness is its hydrogen sulphide (H₂S) component. On Imperial Oil's DilBit Material Safety Data Sheet (MSDS), under the heading *Typical Physical and Chemical Properties*, it states: "'Tarry' odour and associated smell of 'rotten eggs' due to hydrogen sulphide presence" (Imperial Oil, 2002). Enbridge's condensate specs (Enbridge, 2010) list a maximum allowable H₂S concentration as 20ppm.

2.3 Heavy Metal Content

Nickel and vanadium are typically the heavy metals noted when comparing concentrations within conventional crudes and diluted bitumen. Been (2011) notes that vanadium levels are "markedly higher" for the dilbit crudes as compared to the conventional crude oils, while the nickel levels are more comparable with the conventional heavy sour crude oil levels.

The 2010 Baker Hughes white paper compares Athabasca Bitumen to Arab Heavy crude oil and states that nickel, vanadium, nitrogen and asphaltenes are all also significantly higher in the bitumen. Jacobs Canada's 2005 presentation shows a nickel and vanadium content of 61.6ppm and 159.3ppm respectively in dilbit as compared to 3 ppm for each metal in West Texas Intermediate crude. Environment Canada's Oil Properties database lists Athabasca bitumen's nickel range between 69ppm and 85.2ppm, and vanadium range between 81ppm and 218ppm.

2.4 Total Acid Number (TAN)

The Total Acid Number (TAN) is higher in diluted bitumen than in conventional crude oil. TAN is listed in units of mgKOH/g because it is found by titrating potassium hydroxide. The 2006 California Energy Commission report on crude oil imports states that a TAN number greater than 0.5 is considered high.

In Figure 1a of Been (2011), the TAN content is shown to be significantly higher in dilbits. It ranges between 1.0 -1.8 (approximately) for dilbits and between 0.2-0.4 for medium crude oils. In Crandall (2002), it states that many bitumens have high TAN content and lists TAN at 3.5 for Athabasca bitumen and 2.4 for Athabasca dilbit.

2.5 Asphaltenes

Asphaltenes are the components of crude oils and bitumens which have the highest molar mass. They are polyaromatic compounds that have boiling points higher than 500°C, according to Ancheyta *et al.* (2010), who also say: "Asphaltenes are known to be the major precursors of sludge or sediments."

Environment Canada data sheets list asphaltene concentrations of 6% for synthetic dilbit blend and 8.6% for dilbit (EC, 2011). Enbridge's condensate specifications (Enbridge, 2010) list a maximum allowable total suspended solids (TSS) concentration as 200mg/L, while the typical value is 110mg/L.

3. Corrosion and Erosion due to Crude Oils

Crude oil, conventional or otherwise, contains impurities; as such, it is toxic to the environment and corrosive to pipelines, storage tanks and cargo holds. The chemical differences (due to impurities) between diluted bitumen and conventional crude oils in terms of sulphur, heavy metals (predominantly

nickel and vanadium) and total acid number (TAN) only serve to heighten the corrosiveness to metal and to increase toxicity to the environment in the event of a spill (Crandall, 2002; COQA, 2009; NPRA, 2008).

The known corrosive nature of crudes is addressed by the American Society of Testing and Materials' (ASTM) specifically dedicated standard G205-10 *Standard Guide for Determining Corrosivity of Crude Oils*. ASTM's website refers to G205-10 by saying: "This guide is intended to assist in the selection of methodologies that can be used for determining the corrosiveness of crude oil under conditions in which water is present in the liquid state (typically up to 100°C). These conditions normally occur during oil and gas production, storage, and transportation in the pipelines." (ASTM G205-10, 2011)

Li *et al.* (2007) say: "It is well known that the CO₂ corrosion of carbon steel is a key problem in gas _ [sic] oil industry." The nature and type of corrosion, however, is not always easy to determine. Zhu *et al.* (2) (2010) note: "Although an abundance of research results has been published, the mechanism of localized pitting corrosion in sour environment is not well understood yet." In Ayello *et al.*'s 2008 study of crude oil chemistry effects on corrosion rates, it states: "the molecular composition of crude oil is very complex and crude oil chemistry varies from field to field. It is therefore impossible to study the effect of each and every chemical present in crude oil."

The National Research Council of Canada (NRC) website on pipeline integrity states: "From 1980-1997, a total of 12 137 operating pipeline failures were recorded, averaging 674 failures per year. Internal corrosion caused the majority (about 50%) of these failures." Of these internal corrosion failures listed by the NRC, pipelines carrying crude oil are included. Sand *et al.* (2005) add, "Internal corrosion affects millions of miles/kilometres of pipelines everyday resulting in failures, lost production and, in some cases, loss of lives."

3.1 Corrosiveness of Crude Oils due to Sulphur Content

Average sulphur concentrations in diluted bitumen are eight times higher than in average benchmark crude oils (COQA, 2009; Environment Canada website). As a naturally-occurring carbon compound, bitumen also contains sulphate reducing bacteria (SRB). These organisms can live and reproduce in a variety of pH, temperature and pressure environments, including petroleum products. Type cultures of SRB have been obtained from oilfield production facilities and from Athabasca oil sands (Armstrong *et al.*, 1998). Microorganisms can participate in the corrosion process by consuming cathodic hydrogen from the steel or by producing corrosive metabolic end products such as sulphide. SRB can influence steel corrosion in nominally aerated environments, as well as in anaerobic environments (Garverick, 1994; Pankhania, 1988). SRB metabolise organic matter and dissolved sulphate in an anoxic environment to create H₂S and bicarbonate (HCO₃).

Steel is an alloy susceptible to reaction with hydrogen sulphide (H₂S). Hydrogen sulphide then reacts with water to disassociate to form H ions. Atomic hydrogen (H) is capable of diffusing through steel whereas the larger molecular form, H₂, is not. The atomic form of hydrogen diffuses into metal voids where it combines into hydrogen gas which becomes trapped. Atomic hydrogen can also migrate into the microstructure of the metal, or adsorb to the metal surface. Excess hydrogen can lead to the many classes of damage including stress corrosion cracking, hydrogen-assisted cracking and corrosion fatigue.

Dissolved free sulphides (H_2S , HS^- and S_2^{2-}) present in aqueous solution can be responsible for stress cracking (Mostert *et al.*, 2005). Numerous scientific studies agree that hydrogen accumulation and its effects are arguably the predominant concern for buried steel pipeline integrity (Elboudjaini *et al.*, 2004; Craig *et al.*, 2005; Kim *et al.*, 2002; Mostert *et al.*, 2005).

As well, the National Research Council website states “Hydrogen-induced cracking (HIC) is caused by nascent hydrogen atoms (H^0), usually produced in aqueous hydrogen sulphide (H_2S)... Hydrogen atoms that enter the pipeline can cause embrittlement and failure.”

3.2 Further Risks Resulting from Corrosive Environments

Beyond the obvious threats to pipeline integrity due to corrosion, stress corrosion cracking (SCC) is also a result from a steel pipeline being subjected simultaneously to a tensile stress and a corrosive environment. SCC used to be seen in the pipeline industry as an external problem (resulting from a corrosive soil-to-pipe environment) but is now recognized as occurring within pipelines as well. Eliassen *et al.* (2000) note in their Corrosion 2000 NACE Int. Conference paper that: “Increased attention to the problem of SCC has arisen during the last few years, particularly related to a number of incidents in Canada since 1985.”

3.3 Total Acid Number (TAN)

While a 2006 paper from the California Energy Commission points out that, “some acids are relatively inert. Thus, the TAN number does not always represent the corrosive properties of the crude oil.”, the same paper also states: “...high TAN crude oils contain naphthenic acids, a broad group of organic acids that are usually composed of carboxylic acid compounds. These acids corrode the distillation unit in the refinery and form sludge and gum which can block pipelines and pumps entering the refinery.” Pipeline operating temperatures, however, are low enough that corrosiveness of a high TAN content is not necessarily of primary concern. Instead, the primary concern with TAN is the known toxicity of a high TAN-content crude or diluted bitumen. Imperial Oil’s MSDS for diluted bitumen, under the heading *Toxicity Data*, states: “Not available for product”. It is undisputed, however, that crude oils contain TANs and bitumen and dilbits contain even higher TAN concentrations. In Maxam’s 2010 review of naphthenic acid (the primary component in TAN), it is noted that naphthenic acids are persistent in the environment and “NA’s are recognized as one of the most toxic components in oil sands process waters..Likely mode of toxicity is narcosis”.

3.4 Erosion Risk from Particulates

As stated in a paper presented at a 2010 NACE Int. conference:

“The presence of sand in petroleum pipelines has presented a range of challenges to the petroleum industry. When solids travel in a pipeline system, they increase pressure drawdown...and cause the erosion of equipment...Furthermore, the deposited sands can trigger under-deposit pitting corrosion, leading to eventual pipeline failures.”

Zhu *et al.* (1),2010

Asphaltenes can precipitate out of diluted bitumen and be problematic in pipelines. Ancheyta *et al.* (2010) state: “Asphaltenes are known to be the major precursors of sludge or sediments.” Been (2011) says that depending on the particular diluent characteristics, its mixing with bitumen could “result in the

formation of unstable asphaltenes micro-emulsions that could deposit during pipeline transportation". Beck *et al.* (2005) add that asphaltenes can precipitate from changes in pressure, temperature or composition "and can be a major problem for oil producers...Asphaltene deposition in surface facilities and pipelines can occur upon ... the addition of condensate diluent".

Ajmera *et al.* (2010) discuss research into asphaltenes acting as corrosion inhibitors in pipelines by adsorbing onto metal surfaces, thereby creating a protective barrier. However, the actuality of what is occurring has not been verified; scientists agree that crude oils are so unique from one another that the exact role, if any, of asphaltenes in inhibition is uncertain. Furthermore, the authors state: "Asphaltene deposition in well bores and flowlines is a major concern for flow assurance." (Ajmera *et al.*, 2010)

3.5 Heavy Metals

Been (2011), after noting that the concentrations of nickel and vanadium are higher for dilbits (the vanadium levels markedly so), states that "these metals have not been linked to corrosive processes in oil transmission pipelines". While the corrosiveness of these two heavy metals is unknown, their toxicity in the environment is known.

4 Evidence of Corrosion due to Crude Oil and Diluted Bitumen

The greatest bitumen pipelining activity remains in Alberta at present. A comparison between Alberta's hazardous liquid pipeline system and that in the entire U.S. reveals four times the number of incidents in Alberta from 1990 to 2005 (Alberta Energy and Utilities Board (AEUB), 2007; PHMSA, 2011). The Alberta report lists 81,917 km of hazardous liquid pipeline as having a total of 5,333 incidents over a time span of sixteen years. Over the same time period, the PHMSA recorded 3,763 incidents over a longer network of hazardous pipeline (over 289,620 km according to a 2004 US Congressional Research Service report). When compared over a similar distance, the numbers show that Alberta reported quadruple the number of incidents per kilometre.

The average age of pipeline in Alberta is much less than the average age of U.S. pipeline. The majority of Alberta's hazardous liquid pipeline is less than twenty years old (AEUB, 2007), while the majority of pipeline in the U.S. was installed prior to 1970 (Christian Science Monitor, 2010).

The 2007 AEUB report also shows that internal corrosion accounted for 49% of reported pipeline incidents in Alberta's hazardous liquid pipeline system. Internal corrosion caused only 8.5% of incidents in the U.S. hazardous liquid pipeline system over the same time period (Swift and Shope, 2010). When the numbers are standardized, Alberta reported twenty-five times the number of internal corrosion incidents per kilometre of pipeline than the U.S. system. According to Sand *et al.* (2005), internal corrosion has accounted for 53% of all pipeline failures in Alberta between 1980 and 2002.

Been (2011) addresses this 2007 AEUB report: "The data shows that the internal corrosion rates in Alberta and in the U.S. are comparable, which indicates that there is no evidence that dilbit would be more corrosive than conventional crudes." The undisputed fact that Alberta's system is much newer, however, means that corrosion is still occurring at a higher rate in the system which carries more diluted bitumen than in the system which carries more crude oil.

In Enbridge's diluted bitumen spill of July, 2010 in Michigan, the company identified hydrogen sulphide (H_2S) as a risk for its personnel on site (Enbridge, 2010). Furthermore, its Health and Safety Plan included Material Safety Data Sheets (MSDSs) on diluted bitumen as well as benzene, hexanes and H_2S . Though final reports on this spill have not yet been released, evidence of corrosion was noted prior to the spill. Hydrogen sulphide, as previously discussed, is extremely corrosive. In Been (2011), it states: "Although the H_2S concentrations in dilbits are not available, there is no indication that these levels would be higher than in conventional crudes [4]...If available hydrogen sulphide could separate from the oil into an aqueous phase in the pipeline, the corrosiveness of the water would increase."

After Enbridge's July, 2010 diluted bitumen pipeline rupture, the Michigan State public health surveillance system identified 320 individuals with adverse health effects who live along the impacted waterways (Stanbury, 2010). These adverse effects were disclosed through health care provider reports. The Michigan Department of Community Health also issued a fish advisory, which included fish of any kind downstream of the spill not being safe for contact or consumption. Water and soil sample results from this spill, posted on both the Enbridge and EPA websites, show elevated levels of arsenic, benzene, butane, butylbenzyl phthalate, ethylbenzene, iron, nickel, PAHs, sulphur, toluene, vanadium and xylene, amongst others.

On PHMSA's Significant Incidents Listing website, it lists six Enbridge pipeline spills since 2004 which are due specifically to internal corrosion; the volume from these spills totalled 111,610 L. Internal corrosion was the proven cause of the Prudhoe Bay, Alaska oil pipeline rupture which spilled over 800,000 litres of crude oil (Alaska Department of Environmental Conservation, 2008). A 2004 crude oil pipeline inspection of this pipeline revealed severe internal microbial corrosion (Lilly *et al.*, 2007). A March 2006 oil spill of 803,460 litres at the same site was confirmed by the Department of Conservation final report to have been caused by internal corrosion. Inspection of the pipeline after this leak "showed evidence of high rates of corrosion, even in places that has been free of corrosion in the fall 2005 inspection" as quoted from Petroleum News in 2006.

6. Conclusion

The pipelining of diluted bitumen as opposed to conventional crude oil is relatively new to the oil industry, and more research is required to determine how much more corrosion damage, if any, is occurring. However, all sour crude oils are corrosive and toxic to the environment in the event of a spill.

The 2007 Alberta Energy and Utilities Board report and Enbridge's 2010 diluted bitumen spill in Michigan are only two examples which require further investigation. Studies have been commissioned in both Canada and the US on the effects of diluted bitumen on pipelines, indicating that more information is necessary for safety and risk assessments. Until such a time as these studies are available, any proposed projects involving diluted bitumen have too many uncertainties and therefore pose unknown risks.

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